

The forward problem: hyperelastic membrane inflation

a constitutive ground truth for closed-surface tension (method M3)

Working notes (forward neo-Hookean membrane FE)

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1 Why a forward problem, and what it gives us

The tension-inference pipeline (companion notes, `tension_inference.tex`) solves the *inverse* membrane problem: given a deformed shape Γ and a pressure Δp , find the surface stress that balances them. On a closed surface this balance is statically determinate in principle but its discretisation carries a self-equilibrated null space, so the inverse solve is ill-posed and must be regularised. There is, consequently, no *exact* reference stress on a general closed shape against which to measure the inferred field — only analytic solutions for the sphere, spheroid and capsule.

The *forward* problem removes that gap. We posit a stress-free reference surface, a hyperelastic constitutive law and a pressure, and solve for the deformed configuration and its stress by energy minimisation. Because the elastic energy is convex for moderate strains, the forward solution is

unique: the constitutive law and the reference together *select one member* of the self-equilibrated family that makes the inverse problem ambiguous. The forward stress is therefore a clean, null-mode-free ground truth — including the deviatoric and minor-principal components that the inverse solve recovers least well — and it is available on *arbitrary* shapes, not just the analytic ones. This is method M3 of the project matrix.

The key modelling choice (why the shape need not be prescribed exactly). We are interested in the *tension distribution*, not the precise deformed shape. On a pressurised membrane the tension is governed by shape and pressure (quasi-determinate), so it is insensitive to small shape changes. We exploit this: take the *target* shape as the (near) stress-free reference, choose a *stiff* material, and apply the pressure. The deformed shape then drifts from the target only by the elastic strain $\varepsilon \sim \sigma / (\text{membrane stiffness})$, which we make small ($\lesssim 1\%$) by choosing the shear modulus μ large. The recovered tension is then the equilibrium tension on (essentially) the target geometry. As μ grows the strain vanishes but the stress stays finite (it is fixed by equilibrium), so the limit is well behaved; μ is kept finite only so that the constitutive law genuinely picks the unique solution.

2 Kinematics of a surface membrane

Let the reference (stress-free) midsurface be $\Gamma_0 \subset \mathbb{R}^3$ with material coordinates $\boldsymbol{\xi} = (\xi^1, \xi^2)$ and embedding $\mathbf{X}(\boldsymbol{\xi})$. Deformation maps it to Γ with $\mathbf{x}(\boldsymbol{\xi})$. The reference and deformed covariant tangent bases are

$$\mathbf{A}_\alpha = \frac{\partial \mathbf{X}}{\partial \xi^\alpha}, \quad \mathbf{a}_\alpha = \frac{\partial \mathbf{x}}{\partial \xi^\alpha}, \quad \alpha \in \{1, 2\}. \quad (1)$$

The *surface deformation gradient* is the (3×2) two-point tensor

$$\mathbf{F} = \mathbf{a}_\alpha \otimes \mathbf{A}^\alpha, \quad \mathbf{A}^\alpha \cdot \mathbf{A}_\beta = \delta_\beta^\alpha, \quad (2)$$

mapping reference tangent vectors to deformed ones. The (2×2) , in-plane) right Cauchy–Green tensor and its invariants are

$$\mathbf{C} = \mathbf{F}^\top \mathbf{F}, \quad C_{\alpha\beta} = \mathbf{a}_\alpha \cdot \mathbf{a}_\beta, \quad I = \text{tr}_A \mathbf{C} = A^{\alpha\beta} C_{\alpha\beta}, \quad J = \sqrt{\det(A^{\alpha\gamma} C_{\gamma\beta})}, \quad (3)$$

where $A^{\alpha\beta}$ is the inverse reference metric. I is the trace of the in-plane stretch ($I = \lambda_1^2 + \lambda_2^2$ in principal stretches λ_i) and $J = \lambda_1 \lambda_2$ is the local *areal* stretch (deformed area / reference area). For a flat, isotropically-meshed reference one may take $A_{\alpha\beta} = \delta_{\alpha\beta}$ per element (Section 5).

3 Constitutive law: 2D membrane neo-Hookean

We use the same membrane neo-Hookean energy as the cMSM forward simulations (Marín-Llauradó et al., 2023), a strain energy *per unit reference area*

$$\psi(I, J) = \mu (I - 2) + \lambda (J - 1)^2, \quad (4)$$

with shear modulus μ and an areal-penalty modulus λ (their choice $\lambda = 2\mu$; both have units of surface tension, N/m). The $\mu(I - 2)$ term resists shear/stretch; the $\lambda(J - 1)^2$ term resists area

change and supplies a stable equilibrium under pressure. The second Piola–Kirchhoff surface stress follows from $\mathbf{S} = 2 \partial\psi/\partial\mathbf{C}$:

$$\mathbf{S} = 2 \frac{\partial\psi}{\partial I} \mathbf{A}^{-1} + 2 \frac{\partial\psi}{\partial J} \frac{\partial J}{\partial \mathbf{C}} = 2\mu \mathbf{A}^{-1} + \lambda(J-1) J \mathbf{C}^{-1}, \quad (5)$$

using $\partial I/\partial \mathbf{C} = \mathbf{A}^{-1}$ (reference inverse metric) and $\partial J/\partial \mathbf{C} = \frac{1}{2} J \mathbf{C}^{-1}$. The physical *Cauchy membrane stress* (force per unit deformed length, i.e. the surface tension tensor compared against the inference) is the push-forward

$$\boldsymbol{\sigma} = \frac{1}{J} \mathbf{F} \mathbf{S} \mathbf{F}^\top, \quad (6)$$

a symmetric in-plane (3×3 , rank-2) tensor with $\boldsymbol{\sigma} \mathbf{n} = \mathbf{0}$. Its non-zero eigenvalues are the principal tensions σ_1, σ_2 that we compare to the inferred field. (Note σ here is a force-per-length *resultant*, matching our N ; divide by thickness t if a true stress is wanted.)

4 Equilibrium: energy functional and weak form

4.1 Total potential energy

The deformed configuration minimises the total potential energy

$$\Pi[\mathbf{x}] = \underbrace{\int_{\Gamma_0} \psi(I, J) dA_0}_{\text{elastic energy}} - \underbrace{\Delta p V[\mathbf{x}]}_{\text{pressure work}}, \quad V[\mathbf{x}] = \frac{1}{3} \oint_{\Gamma} \mathbf{x} \cdot \mathbf{n} da, \quad (7)$$

where V is the volume enclosed by the closed deformed surface (divergence theorem) and Δp is the internal–external pressure difference. Stationarity $\delta\Pi = 0$ for all admissible variations $\delta\mathbf{x}$ gives the weak form

$$\underbrace{\int_{\Gamma_0} \mathbf{S} : \delta \mathbf{E} dA_0}_{\text{internal virtual work}} = \underbrace{\Delta p \oint_{\Gamma} (\mathbf{n} \cdot \delta \mathbf{x}) da}_{\text{external (pressure) virtual work}}, \quad \delta \mathbf{E} = \frac{1}{2} (\mathbf{F}^\top \delta \mathbf{F} + \delta \mathbf{F}^\top \mathbf{F}), \quad (8)$$

$\mathbf{E} = \frac{1}{2}(\mathbf{C} - \mathbf{A})$ being the Green–Lagrange strain. The right-hand side is a *follower* load: the normal $\mathbf{n}$ and area element da are evaluated on the *deformed* surface, so the pressure term depends on the unknown $\mathbf{x}$. Equivalently, the pressure work is simply $\Delta p \delta V = \Delta p \oint \mathbf{n} \cdot \delta \mathbf{x} da$, which is where the follower form comes from.

4.2 Pressure control vs. volume control

Two ways to drive the inflation:

- **Pressure-controlled:** Δp prescribed, V free. Simple, but a closed membrane can be *unstable* once past the inflation limit point (the balloon instability: $d(\Delta p)/dV < 0$), where Newton diverges.
- **Volume-controlled:** prescribe the enclosed volume $V = V^*$ via a Lagrange multiplier; the multiplier *is* the pressure Δp (a readout). The augmented stationarity is $\int \mathbf{S} : \delta \mathbf{E} dA_0 - \Delta p \delta V = 0$, $V[\mathbf{x}] = V^*$. This traverses the limit point stably and is exactly the device the cMSM vertex model used to read off lumen pressure.

For a stiff membrane at modest Δp we are far below the limit point and pressure control is adequate; volume control is the robust fallback.

5 Finite-element discretisation: P1 membrane triangles

The reference surface is a triangulation; the unknowns are the deformed nodal positions $\mathbf{x}_i \in \mathbb{R}^3$ (3 DOF/node). We use the **constant-strain triangle (CST) membrane element**: piecewise-linear (P1) geometry, so each triangle has a *constant* deformation gradient and hence constant stress. This is the standard membrane element and the counterpart of the P1 elements in the inverse solver (`getShapeFunctions.m`, 'Triangle' in the cMSM code).

Element deformation gradient. On a triangle with reference vertices $\mathbf{X}_0, \mathbf{X}_1, \mathbf{X}_2$ and deformed vertices $\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2$, form the edge matrices

$$\mathbf{D}_X = [\mathbf{X}_1 - \mathbf{X}_0 \quad \mathbf{X}_2 - \mathbf{X}_0] \in \mathbb{R}^{3 \times 2}, \quad \mathbf{D}_x = [\mathbf{x}_1 - \mathbf{x}_0 \quad \mathbf{x}_2 - \mathbf{x}_0] \in \mathbb{R}^{3 \times 2}, \quad (9)$$

and a 2D reference frame (orthonormal tangent basis $\mathbf{t}_1, \mathbf{t}_2$ of the reference triangle) so that $\hat{\mathbf{D}}_X = [\mathbf{t}_1 \quad \mathbf{t}_2]^\top \mathbf{D}_X \in \mathbb{R}^{2 \times 2}$ is invertible. Then the surface deformation gradient and Cauchy–Green follow as

$$\mathbf{F} = \mathbf{D}_x \hat{\mathbf{D}}_X^{-1} [\mathbf{t}_1 \quad \mathbf{t}_2]^\top, \quad \mathbf{C} = \mathbf{F}^\top \mathbf{F}, \quad A_0 = \frac{1}{2} |\mathbf{D}_X(:, 1) \times \mathbf{D}_X(:, 2)|, \quad (10)$$

with A_0 the reference triangle area. $I, J, \mathbf{S}, \boldsymbol{\sigma}$ are then constant on the element via (5)–(6).

Element internal force and tangent. The element energy is $\Psi_e = A_0 \psi(I, J)$. Its gradient and Hessian with respect to the 9 element DOFs $\mathbf{u}_e = (\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2)$ give the internal force and material+geometric tangent

$$\mathbf{f}_e^{\text{int}} = \frac{\partial \Psi_e}{\partial \mathbf{u}_e}, \quad \mathbf{K}_e^{\text{mat+geo}} = \frac{\partial^2 \Psi_e}{\partial \mathbf{u}_e \partial \mathbf{u}_e}, \quad (11)$$

assembled into the global residual and stiffness. These derivatives are available in closed form for (4) but are most safely obtained by *automatic differentiation* of $\Psi_e(\mathbf{u}_e)$ (Section 10).

Pressure (follower) load. The discrete enclosed volume and its gradient (the pressure force) are

$$V = \frac{1}{6} \sum_{\text{tri}} \mathbf{x}_0 \cdot (\mathbf{x}_1 \times \mathbf{x}_2), \quad \mathbf{f}^{\text{ext}} = \Delta p \frac{\partial V}{\partial \mathbf{x}}, \quad (12)$$

which distributes the pressure as nodal forces along deformed normals. Its derivative $\partial^2 V / \partial \mathbf{x} \partial \mathbf{x}$ is the *load (pressure) stiffness*; it is non-symmetric, so the global tangent $\mathbf{K} = \mathbf{K}^{\text{mat+geo}} - \Delta p \partial^2 V / \partial \mathbf{x}^2$ is non-symmetric and must be solved with a general (not Cholesky) sparse solver.

6 Boundary conditions: a free-floating pressurised body

The target case is a *closed, free-floating* surface (embryo) under internal pressure — *no* physical boundary, unlike the pinned open domes of cMSM. This is admissible because a closed surface under uniform pressure is *self-equilibrated*: the net pressure force $\oint \Delta p \mathbf{n} da$ and torque vanish. The only obstruction is the 6-dimensional **rigid-body null space** (3 translations + 3 rotations) of the stiffness matrix. Two standard cures:

- **Minimal “3–2–1” restraint**: fix one node in all 3 directions, a second in 2, a third in 1 (6 constraints total, nodes non-collinear). Because the body is in self-equilibrium the reactions are essentially zero, so this introduces *no spurious stress* — it only removes the rigid modes.

- **Rigid-mode projection:** keep all DOFs free and project the 6 rigid modes out of the residual and increment (or use a minimum-norm `lsqr` solve). Equivalent stress, no anchor nodes.

The pressure load is consistent (orthogonal to the rigid modes), so a solution exists. Volume control (Section 4) adds one Lagrange-multiplier row/column to the tangent and replaces Δp by an unknown; the rigid-mode treatment is unchanged.

7 Solution algorithm

1. **Mesh & reference.** Build the reference surface (`vedo.IcoSphere`, the capsule builder, or an embryo mesh); store reference frames, edge matrices $\hat{D}_X$ and areas A_0 once.
2. **Load stepping.** Ramp Δp (or V^*) in n_{step} increments from 0 to the target. For a stiff membrane the response is nearly linear, so $n_{\text{step}} = 1\text{--}5$ suffices; use more only for soft materials or near the limit point.
3. **Newton iteration** per step: assemble residual $\mathbf{r} = \mathbf{f}^{\text{int}} - \Delta p \partial V / \partial \mathbf{x}$ and tangent $\mathbf{K}$; apply the 3–2–1 constraints (or projection); solve $\mathbf{K} \delta \mathbf{u} = -\mathbf{r}$ with a sparse direct solver (`scipy.sparse.linalg.spsolve`) or GMRES for large/non-symmetric systems; update $\mathbf{x} \leftarrow \mathbf{x} + \delta \mathbf{u}$; iterate to $\|\mathbf{r}\| < \text{tol}$ (typically 3–6 iterations).
4. **Stress recovery.** From the converged $\mathbf{F}$ per element compute $\boldsymbol{\sigma}$ (6) and its eigenvalues σ_1, σ_2 ; average to nodes if a nodal field is wanted. Export to VTK alongside the inferred field.
5. **Diagnostics.** Track the shape drift $\max \|\mathbf{x} - \mathbf{X}\|/R$ (should be $\lesssim 1\%$ for the stiff-reference strategy), the enclosed volume, and the smallest tangent eigenvalue (limit-point proximity).

8 Validation plan

Following the project rule (sphere first):

- **Sphere.** A neo-Hookean spherical membrane inflates self-similarly with a closed-form pressure–stretch relation $\Delta p(\lambda)$; check the solver reproduces it and that $\sigma_1 = \sigma_2 = \Delta p R/2$ (resultant). Verify the balloon limit point is *not* crossed at the working Δp .
- **Prolate / oblate spheroid, capsule.** Take the analytic shape as the stress-free reference, inflate with stiff μ , confirm the shape drift is small and the recovered σ_1, σ_2 match the axisymmetric closed forms (Section 1); these become the M3 ground truth.
- **Cross-check against inference.** Feed the (slightly) deformed shape to the inverse FEM and compare — the forward field is the unique, null-mode-free target the inference should recover.

9 Results

The scheme above is implemented in `forward_neohookean.py` (energy minimisation by L-BFGS with the analytic gradient; `numpy/scipy/vedo`). A note on the constitutive form: in practice we use the *stress-free incompressible* neo-Hookean membrane $W(I, J) = \mu_s (I + J^{-2} - 3)$ rather than the cMSM compressible form $\mu(I - 2) + \lambda(J - 1)^2$ of Section 3. The reason is exactly the stiff-reference strategy: the compressible form carries a residual pre-tension 2μ at the reference (its second Piola–Kirchhoff stress is $2\mu \mathbf{A}^{-1} \neq \mathbf{0}$ at $I = 2, J = 1$), which under a stiff modulus would force a large deformation; the incompressible form is genuinely stress-free at the reference ($\mathbf{S} = 2\mu_s(\mathbf{I} - J^{-2}\mathbf{C}^{-1}) = \mathbf{0}$ there), so a stiff μ_s gives the small strain we want.

All runs use IcoSphere subdiv-4 (capsule: $n = 2522$), $\Delta p = 20$ Pa, $\mu_s = 500$ N/m, and the target shape as the stress-free reference. The recovered tensions are compared with the closed-form membrane resultants (Table 1); Figure 1 shows the $\sigma_{\max}$ fields.

Table 1: Forward neo-Hookean inflation vs. the analytic membrane resultants (N/m), at representative locations. “drift” is $\max \|\mathbf{x} - \mathbf{X}\|/\|\mathbf{X}\|$; “belt err” is the median per-triangle relative error of $\sigma_{\max}$ away from the umbilic poles / junction.

shape	location	$\sigma_{\max}$ (analytic)	$\sigma_{\min}$ (analytic)	belt err	drift
sphere	(uniform)	10.02 (10.02)	9.99 (10.02)	0.4%	2.7×10^{-4}
prolate 2:1	equator	17.47 (17.47)	10.02 (9.99)	0.4%	3.2×10^{-3}
capsule	cylinder	19.97 (20.0)	10.02 (10.0)	0.2%	2.5×10^{-3}
capsule	caps	10.05 (10.0)	10.05 (10.0)	0.2%	—
oblate 1:2	equator	10.07 (10.05)	0.0 (−19.70)	6.1%	2.3×10^{-2}

Sphere, prolate, capsule: $\lesssim 0.4\%$. On the three shapes whose analytic field is everywhere tensile the forward solve reproduces the closed form to a few tenths of a percent — the sphere’s isotropy (0.4%), the prolate’s 1.75 equatorial hoop/meridional anisotropy ($\sigma_{\max}/\sigma_{\min} = 1.744$ vs. 1.749), and the capsule’s pressure-vessel field (hoop $\Delta p R$, axial $\Delta p R/2$, isotropic caps). Crucially this is *tighter* than the inverse inference at the same resolution (which sits at 2–7%, `tension_inference.tex` §12), because the forward field carries no null-mode ambiguity: it is the unique ground truth M3 is meant to provide. The shape drift stays $\lesssim 0.3\%$, confirming the stiff-reference strategy.

Oblate: membrane wrinkling under compressive hoop. The oblate is qualitatively different. Its analytic axisymmetric solution demands a strongly *compressive* hoop near the equator ($\sigma_{\min} = -19.7$ N/m), because for a flattened spheroid $r_2 > 2r_1$ there. A *membrane* cannot sustain compression — with no bending stiffness it wrinkles — so the forward solver relaxes the compressive hoop to $\sigma_{\min} \approx 0$ and deforms more to do so (drift 2.3×10^{-2} , an order of magnitude above the other shapes); $\sigma_{\max}$ still tracks the analytic value (10.07 vs. 10.05). This is not a solver defect but real membrane physics, and it is exactly the situation cMSM guard against with their $\det \sigma > 0$ inequality constraint (Section 3 of `tension_inference.tex` §12): a passive membrane projects compressive principal tensions away. (A sustained compressive $\sigma_{\min}$ therefore signals either bending stiffness — a shell, not a membrane — or, in tissue, an *active* pre-tension; this is the active-signature reading developed in the inference notes.) The oblate is thus a useful stress test: it shows the forward model honours the membrane assumption rather than blindly matching a formal solution.

10 Software / packages

The forward problem is genuinely nonlinear (hyperelastic energy + follower load + Newton), unlike the linear inverse solve. Three viable stacks, in order of fit to this project:

(A) Hand-rolled `scipy.sparse` + automatic differentiation (recommended). Consistent with the project’s no-framework convention and reuses the existing `vedo` mesh/IO and VTK export. Needed packages:

- `numpy`, `scipy` — array maths; `scipy.sparse + scipy.sparse.linalg` (`spsolve/gmres`) for the Newton linear solves; `scipy.optimize` optionally for an energy-minimisation formulation (trust-region / L-BFGS as a robust fallback to Newton).
- **Automatic differentiation** of the element energy $\Psi_e(\mathbf{u}_e)$ to get exact internal forces and tangents without hand-deriving (5) and the geometric/pressure stiffness. Either `jax` (`jax.grad`, `jax.hessian`, `vmap` over elements — fastest, vectorised) or `autograd`. This is the single most labour-saving dependency; the CST energy is small (9 DOF/element) so per-element AD is cheap.
- `vedo` (already installed) — reference-mesh construction and VTK output; `numpy`-side assembly into global sparse matrices.

Estimated ~300–500 lines, fully transparent, no Christoffel/shell machinery.

(B) scikit-fem (installed, 12.0.1). Can assemble the forms and apply constraints, and supports custom nonlinear (bi)linear forms with a Newton loop. However, membrane elements *embedded in $\mathbb{R}^3$* (a 2-manifold element carrying 3 nodal DOFs, with the deformed metric as unknown) are not its native setting; the follower-pressure load and the deformed-configuration geometry would still be hand-coded. Usable for the assembly bookkeeping, but it does not remove the hyperelastic-membrane-specific work.

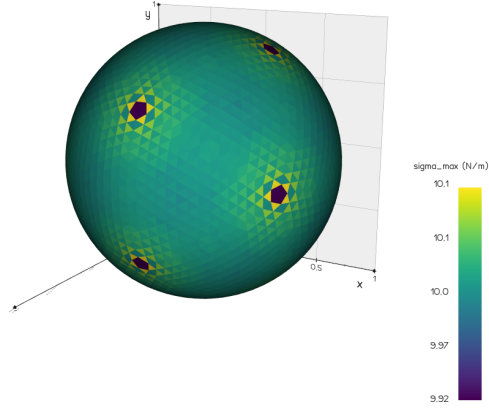
(C) FEniCS/FEniCSx or Firedrake. The most capable for nonlinear hyperelastic shells/membranes: UFL expresses ψ , $\mathbf{F}$, the follower load and the Lagrange-multiplier volume constraint symbolically, with automatic tangents and robust nonlinear solvers (PETSc SNES). The cost is a heavy dependency that is awkward to install on Windows (typically via `conda`/WSL/Docker) and a departure from the project’s lightweight stack. Best reserved for if the hand-rolled solver proves limiting (e.g. large embryo meshes needing PETSc scalability).

Recommendation. Start with (A): `numpy + scipy.sparse` for assembly/solves, `jax` for element-level autodiff of (4), and `vedo` for meshes and VTK. It matches the existing pipeline, keeps the whole forward solver inspectable, and is more than sufficient for sphere/spheroid/capsule and embryo-scale meshes. Escalate to (C) only if PETSc-level performance becomes necessary.

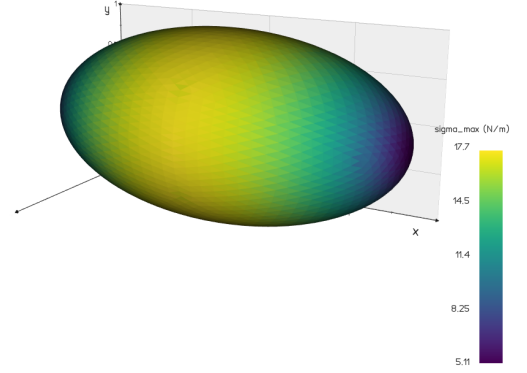
Summary

The forward problem inflates a stress-free reference surface under (follower) internal pressure using a P1 CST membrane neo-Hookean model (4), minimising the total potential energy (7) by Newton iteration with a 3–2–1 (or projection) treatment of the rigid-body modes of a free-floating closed body. Using the target shape as a *stiff* reference keeps the deformed shape within ~ 1% of the target, so the recovered σ_1, σ_2 are the equilibrium tensions on (essentially) the desired geometry — a unique, null-mode-free ground truth (M3) for arbitrary closed surfaces, including embryos with no analytic solution. The recommended implementation is hand-rolled `numpy/scipy.sparse` with `jax` autodiff for the element tangents and `vedo` for mesh I/O.

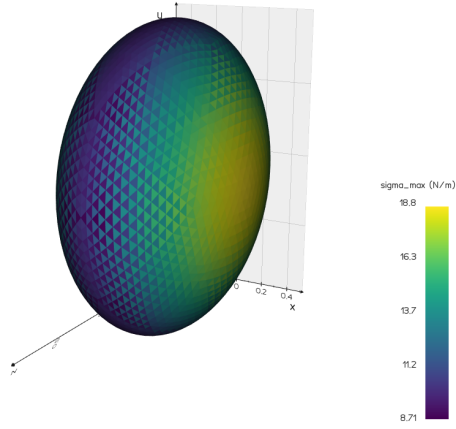
Forward NH sphere | $\sigma_{\max}$ [9.9, 10.1] N/m



Forward NH prolate | $\sigma_{\max}$ [5.1, 17.7] N/m



Forward NH oblate | $\sigma_{\max}$ [8.7, 18.8] N/m



Forward NH capsule | $\sigma_{\max}$ [10.0, 20.0] N/m

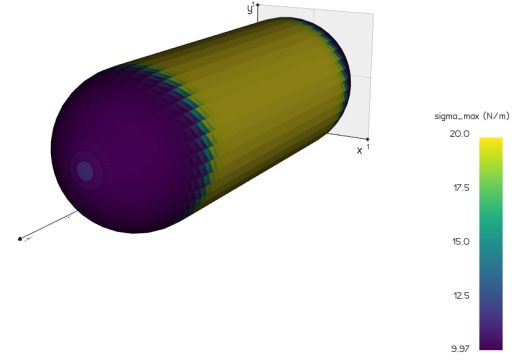


Figure 1: Forward neo-Hookean inflation, deformed surfaces coloured by the larger principal resultant $\sigma_{\max}$ (N/m): sphere (uniform ≈ 10), prolate spheroid (hoop band ≈ 17.5 at the equator fading to ≈ 10 at the poles), oblate spheroid, and capsule (cylinder hoop ≈ 20 , hemispherical caps ≈ 10 with a sharp junction step). The fields are smooth and free of the triangulation “lines” of the inverse solve — the constitutive law selects the unique physical member of the self-equilibrated family.